

Question	Answer	Marks
1(a)	work done per unit mass	B1
	work (done in) moving mass from infinity (to the point)	B1
1(b)(i)	evidence of addition of 3.4×10^6 to 1.7×10^6 or 6.8×10^6	C1
	$GM \times 122 / (5.1 \times 10^6)$ or $GM \times 122 / (10.2 \times 10^6)$	C1
	$6.67 \times 10^{-11} \times M \times 122 \times [(5.1 \times 10^6)^{-1} - (10.2 \times 10^6)^{-1}] = 5.1 \times 10^8$	A1
	leading to $M = 6.4 \times 10^{23}$ kg	
1(b)(ii)	$\phi = (-) (6.67 \times 10^{-11} \times 6.4 \times 10^{23}) / (3.4 \times 10^6)$	C1
	$= -1.3 \times 10^7 \text{ J kg}^{-1}$	A1
1(c)(i)	Mars takes (just under) 25 hours to rotate once on its axis	B1
1(c)(ii)	orbit is equatorial or orbit is in same direction as direction of rotation of Mars	B1

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